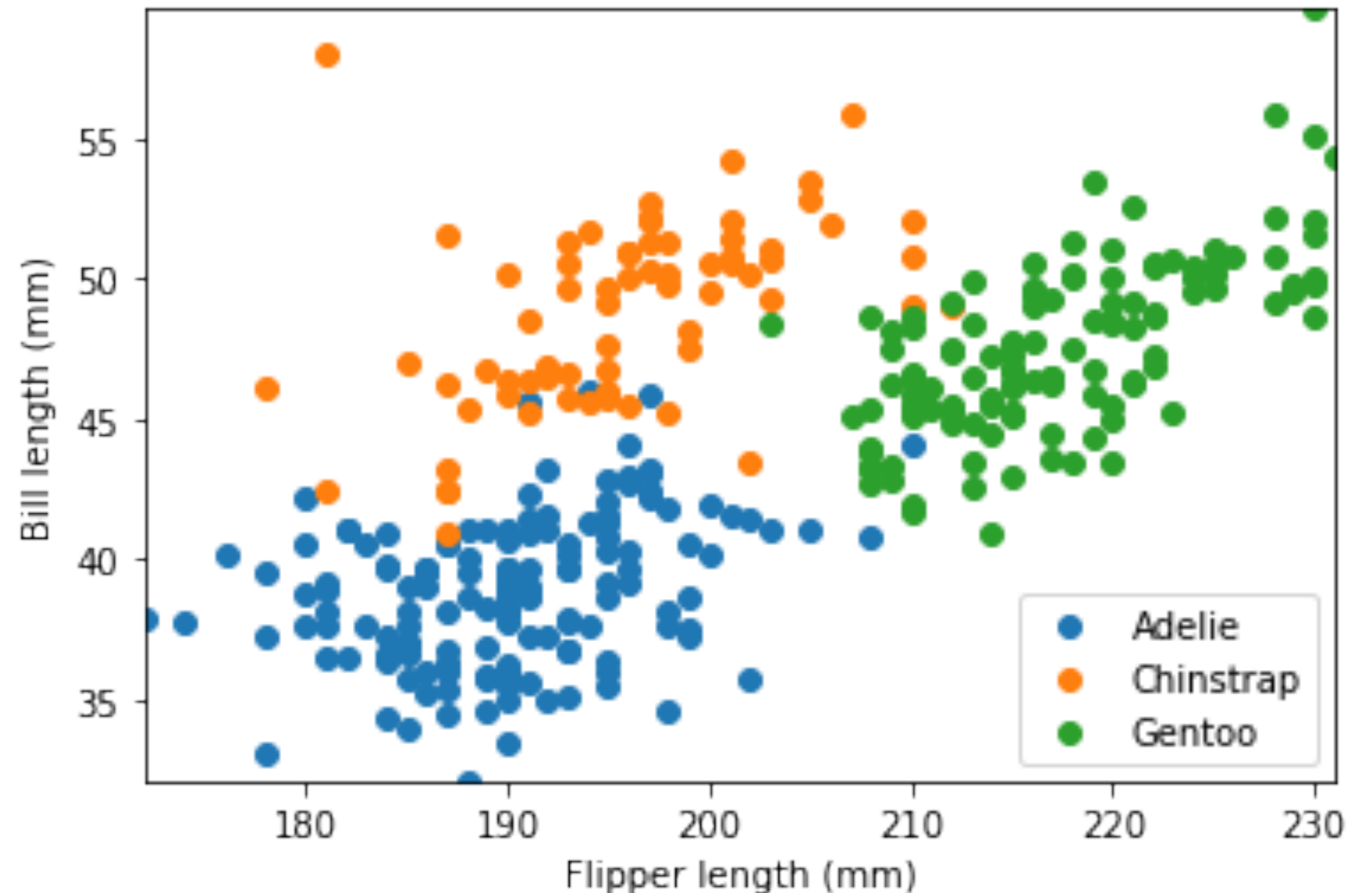


Multi-Class Linear Classifier

CSC 487: Deep Learning
Instructor: Jonathan Ventura

Multi-class linear classifier

- What if we have more than two classes?
- Features: flipper length (x-axis)
bill length (y-axis)
- Label: species (Adelie, Chinstrap, or Gentoo)



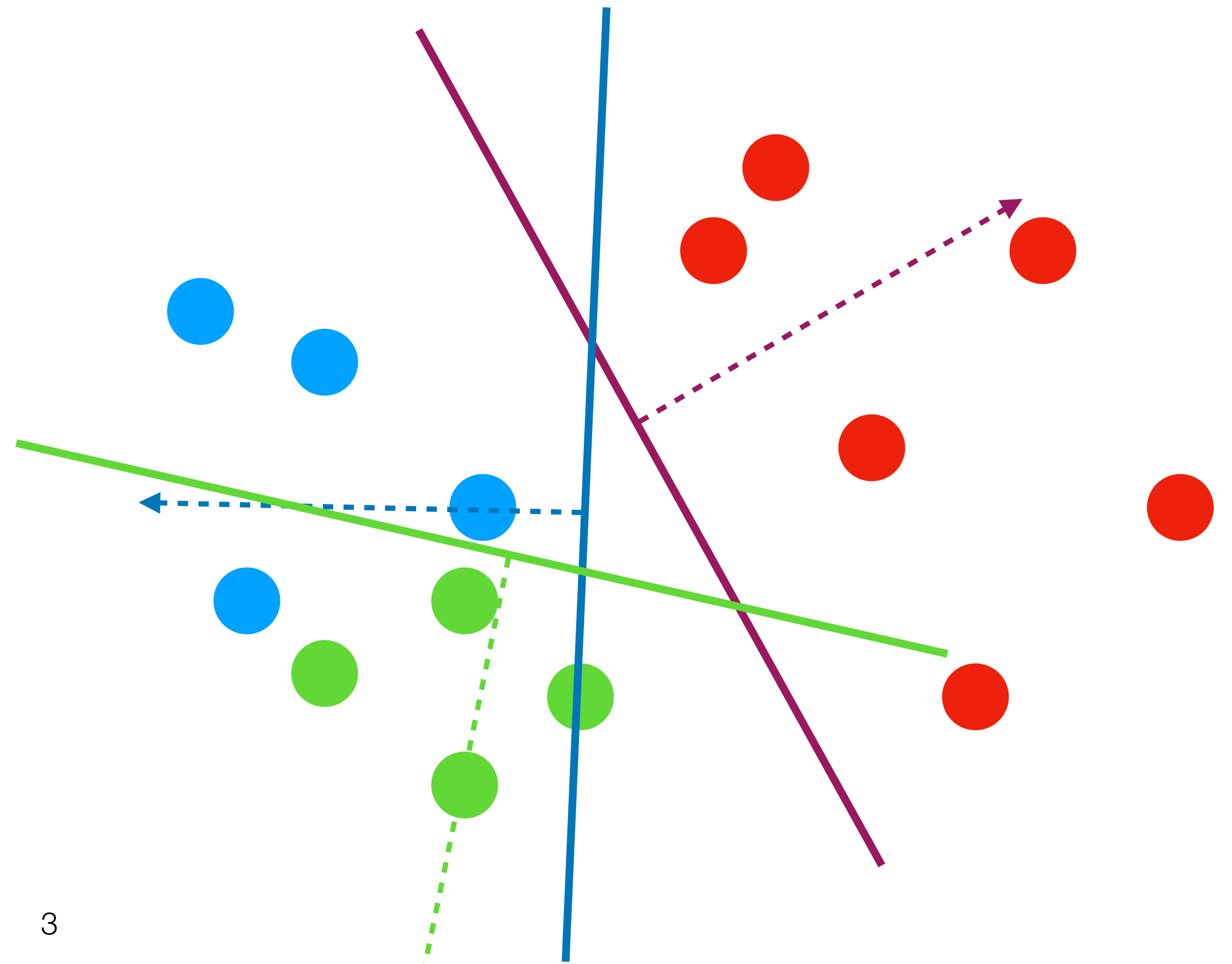
One-vs-rest classification

- A multi-class linear classifier performs one projection per class

$$z_0 = \mathbf{w}_0^T \mathbf{x} + b_0$$

$$z_1 = \mathbf{w}_1^T \mathbf{x} + b_1$$

$$z_2 = \mathbf{w}_2^T \mathbf{x} + b_2$$



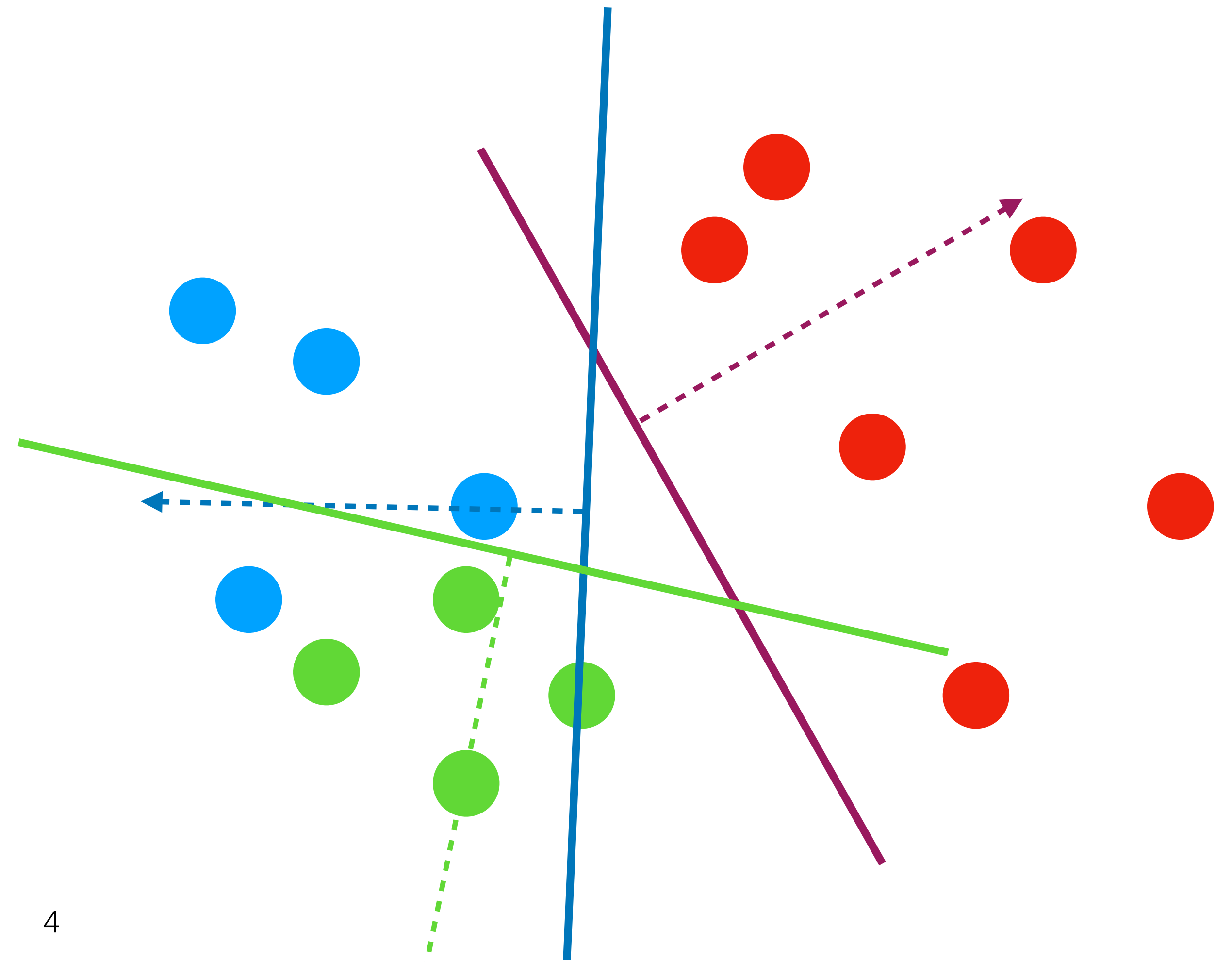
Each point assigned to class with greatest z value

$$z_0 = \mathbf{w}_0^T \mathbf{x} + b_0$$

$$z_1 = \mathbf{w}_1^T \mathbf{x} + b_1$$

$$z_2 = \mathbf{w}_2^T \mathbf{x} + b_2$$

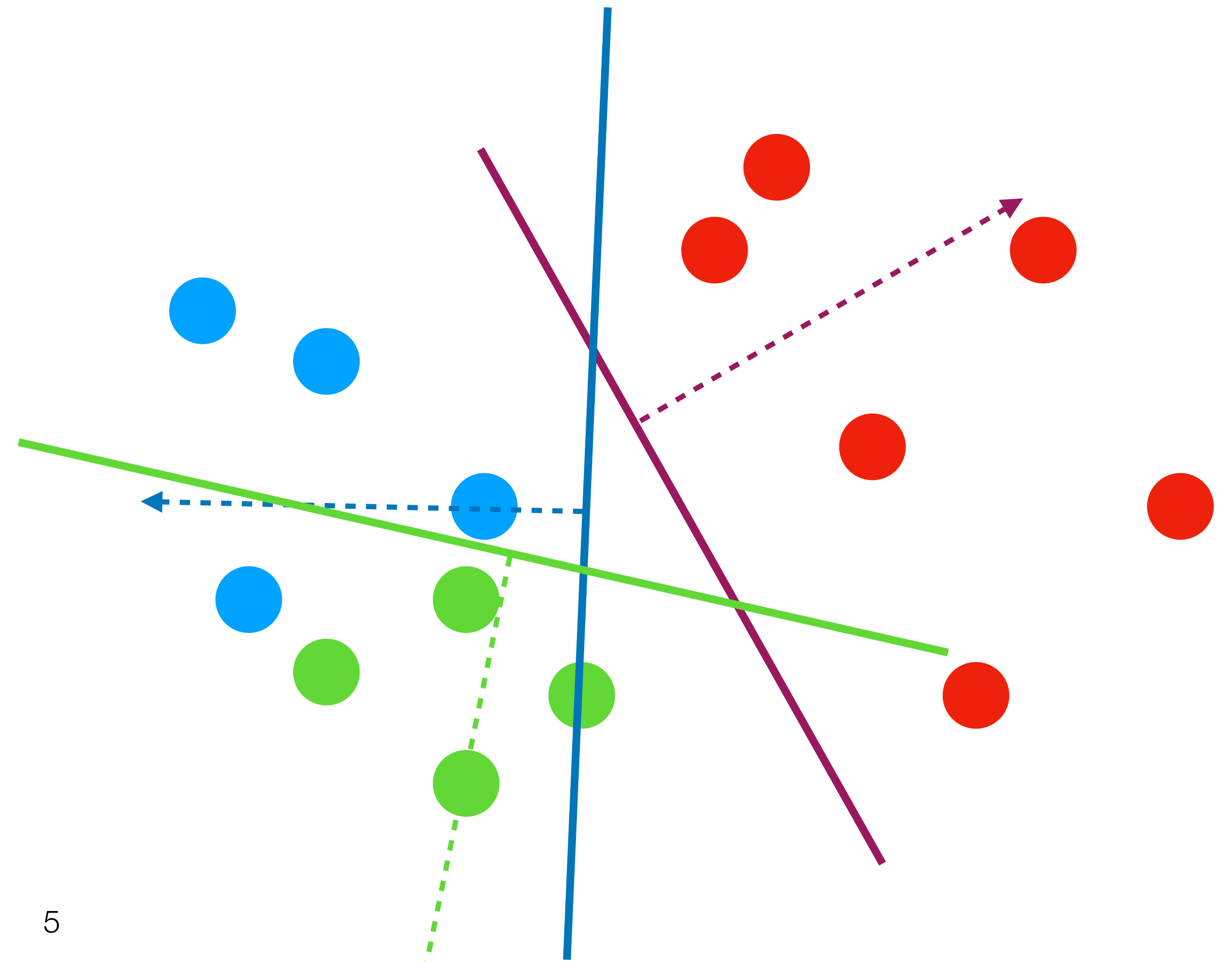
$$y = \mathbf{argmax}_i z_i$$



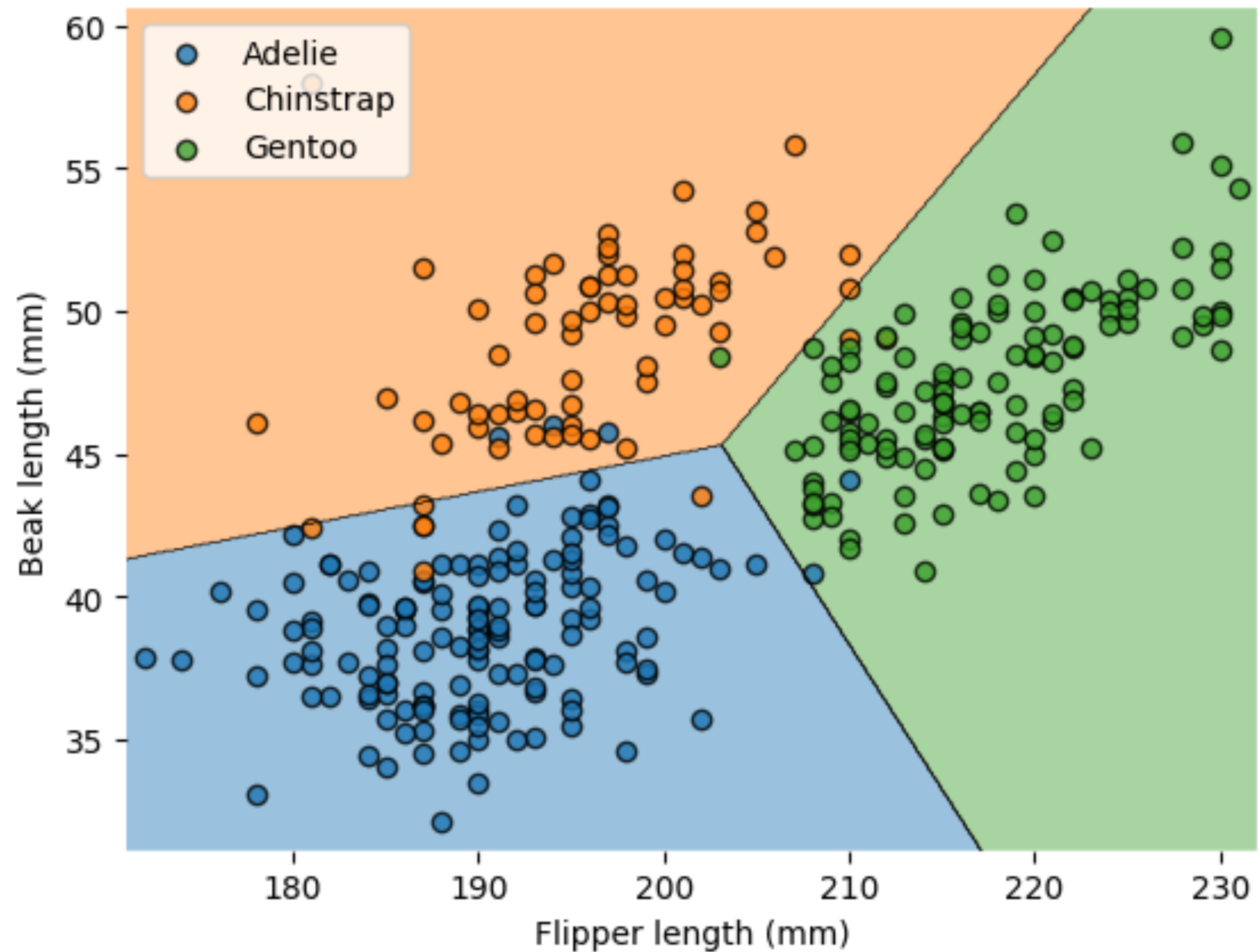
Linear classifier is matrix-vector multiply

- Can combine multiple linear classifiers into a single affine transformation:

$$\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$$

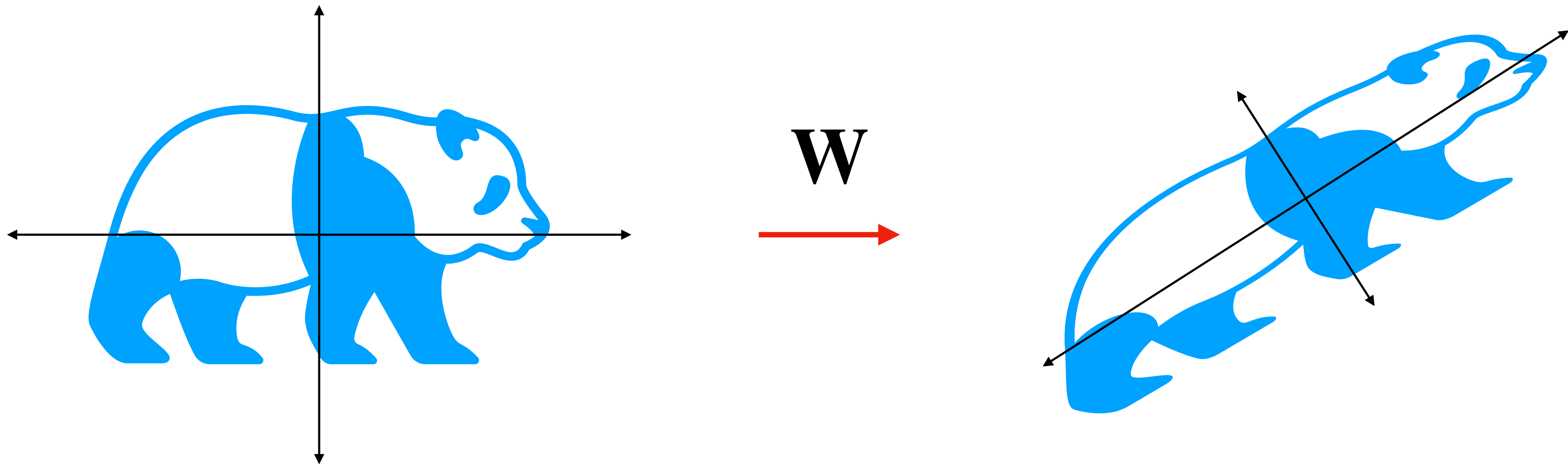


Multi-class classifier result



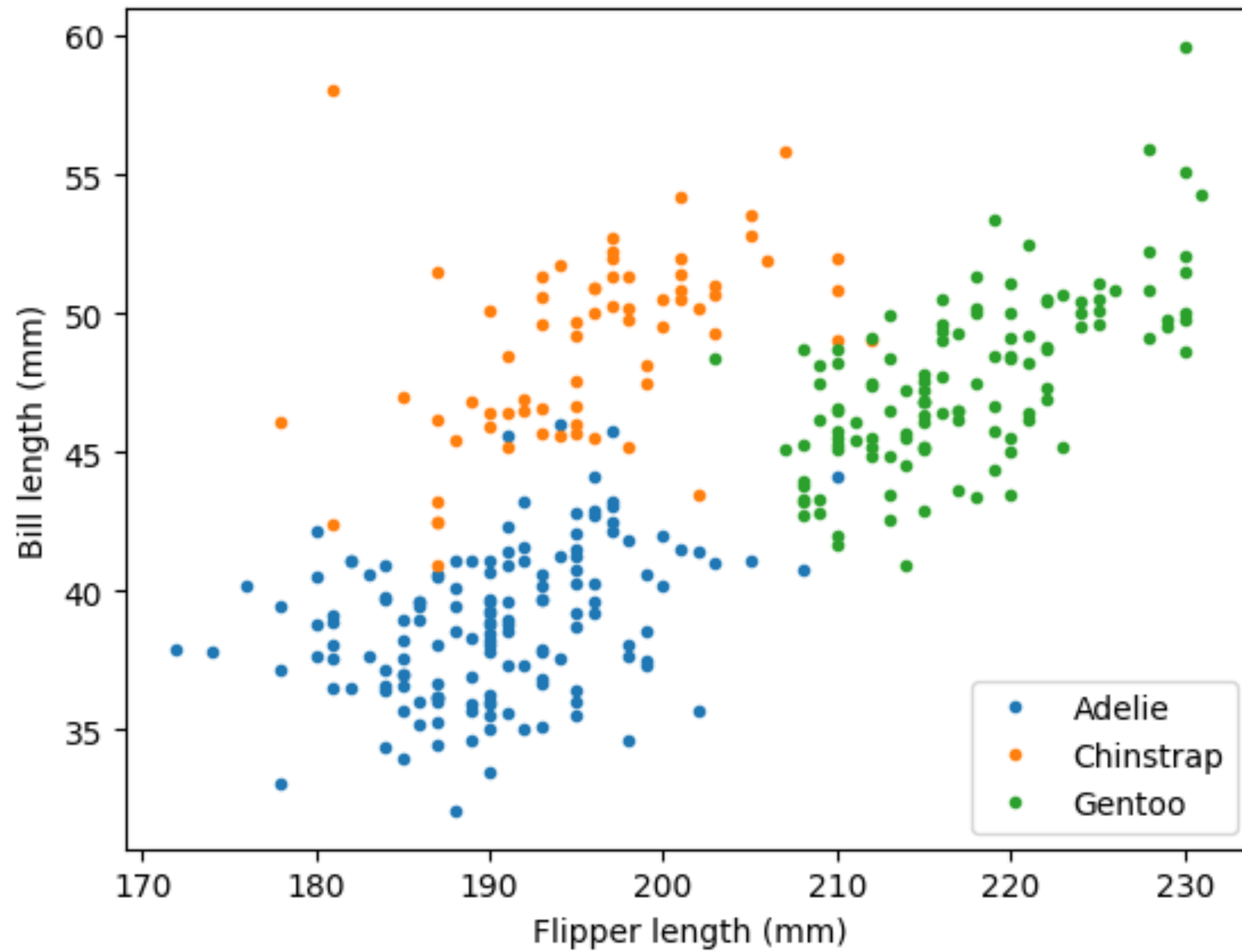
Linear transformation

- Can think of the matrix $\mathbf{W}$ as **transforming** the points $\mathbf{x}$ to a new coordinate system by **rotating**, **translating**, **stretching**, and **shearing**.

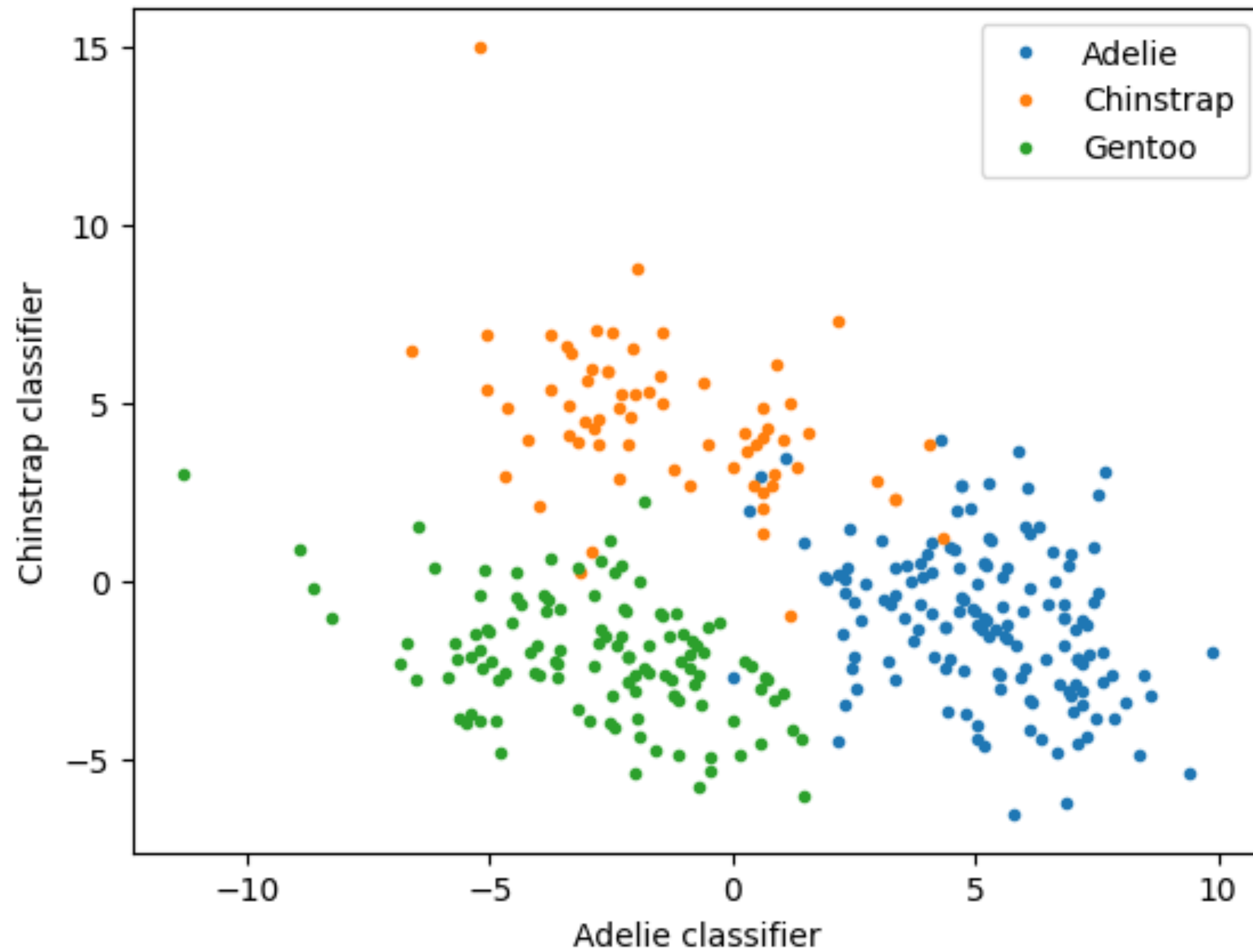


- Can also **add** or **remove** dimensions

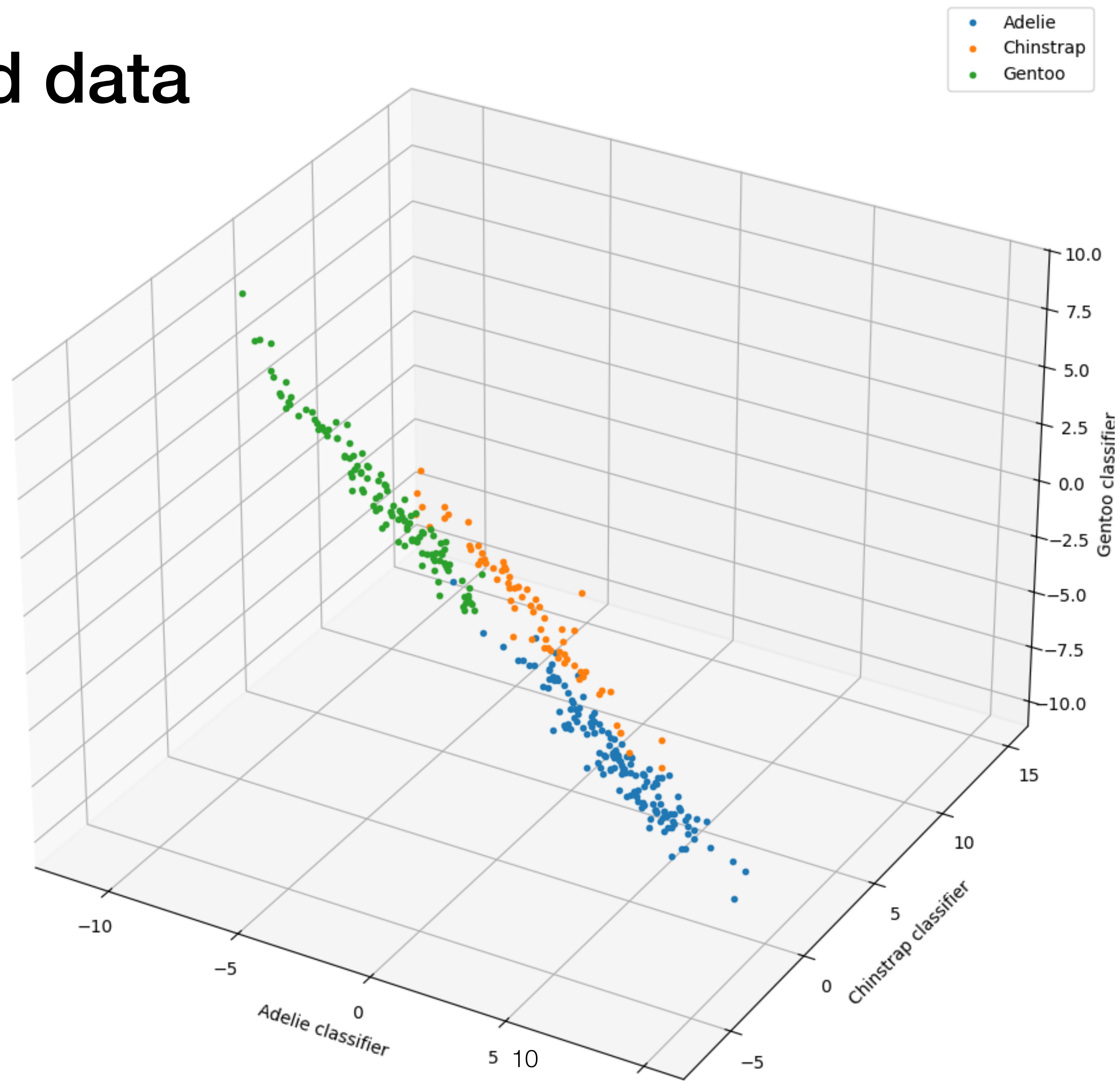
Original data



Transformed data (2D)



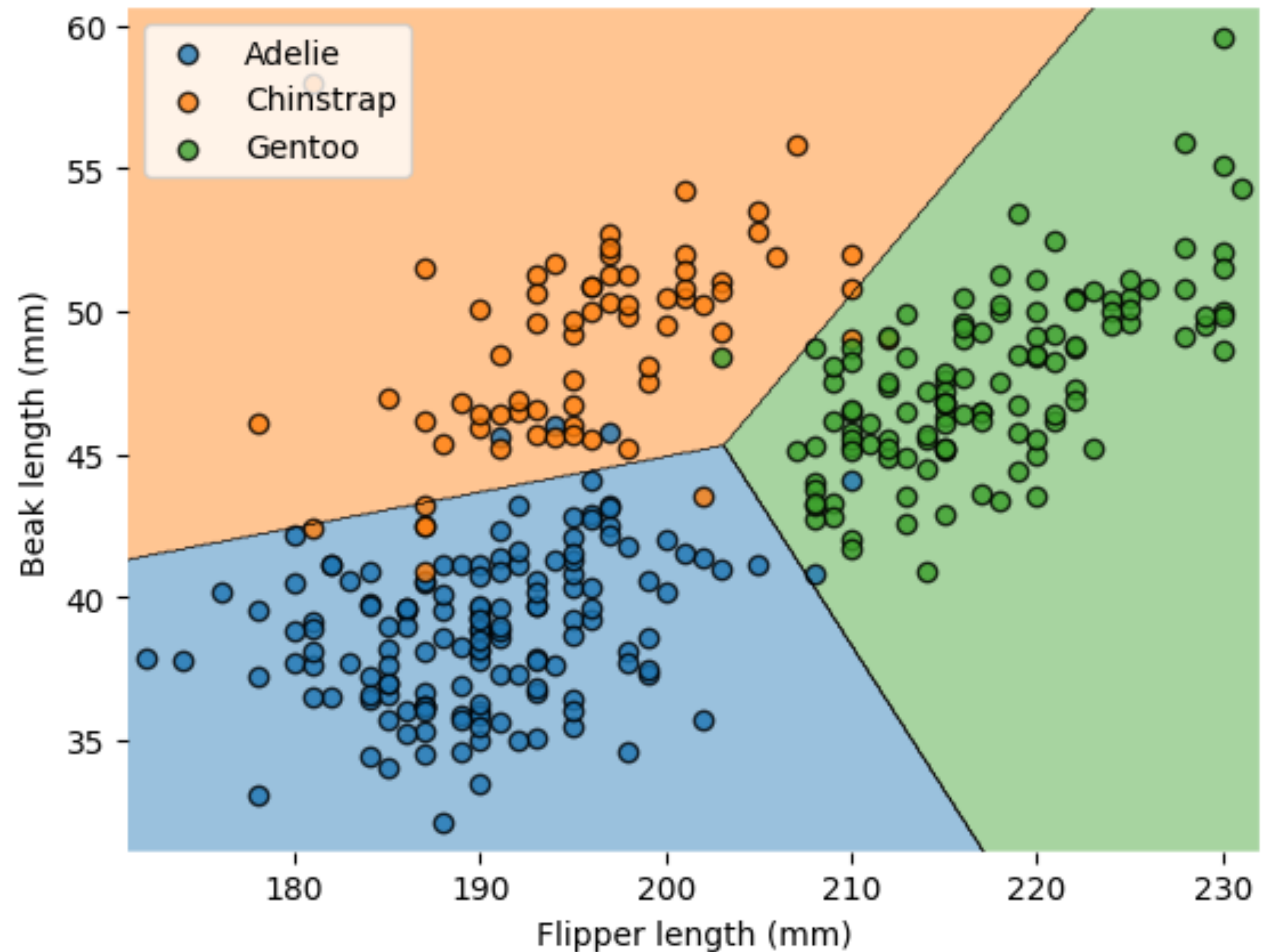
Transformed data in 3D



Evaluating the Model

Evaluating model “fit”

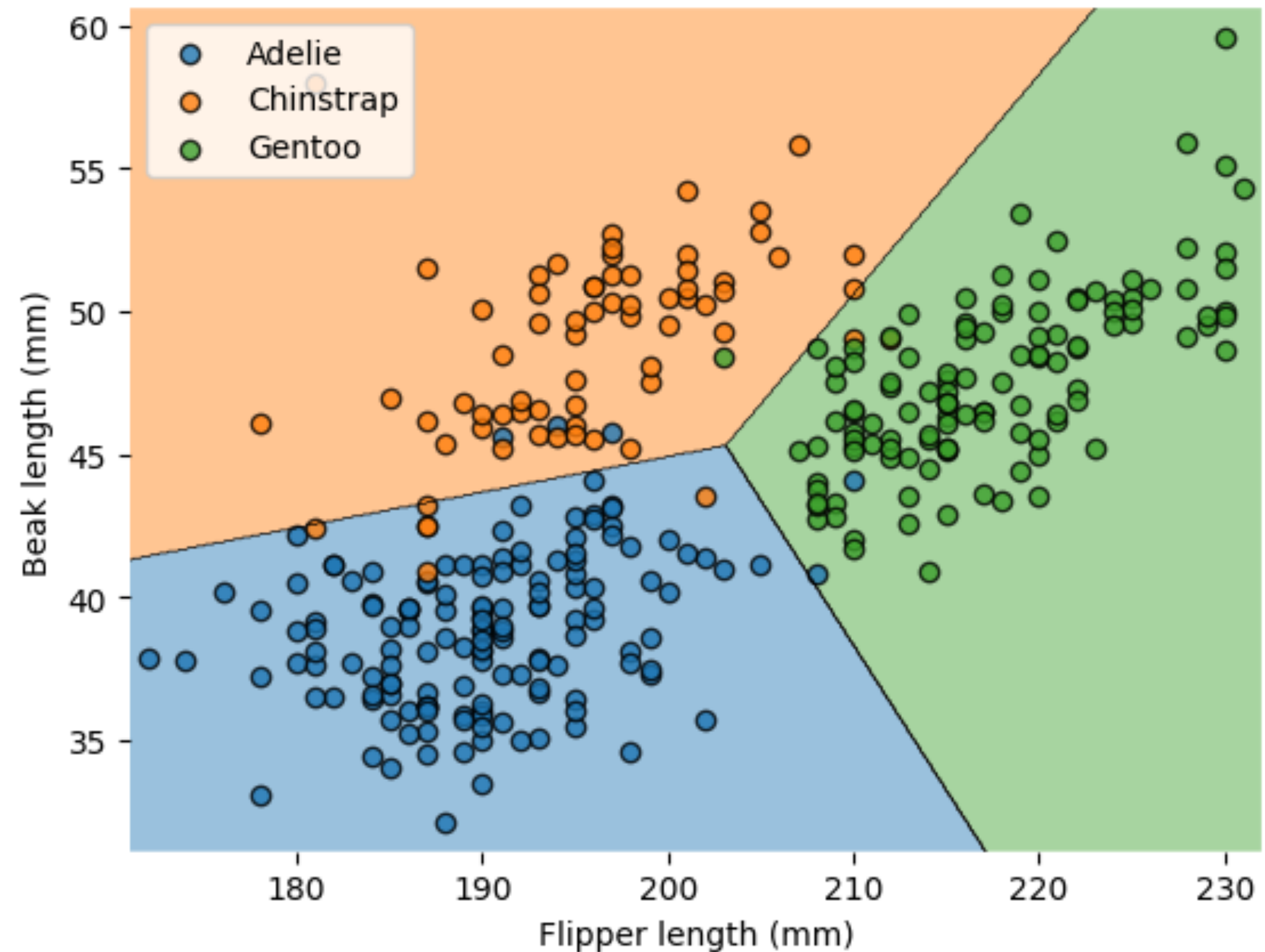
- We often want to evaluate the “fitness” of the model to the data, expressed as a single number.
- How could we compute this?



Accuracy

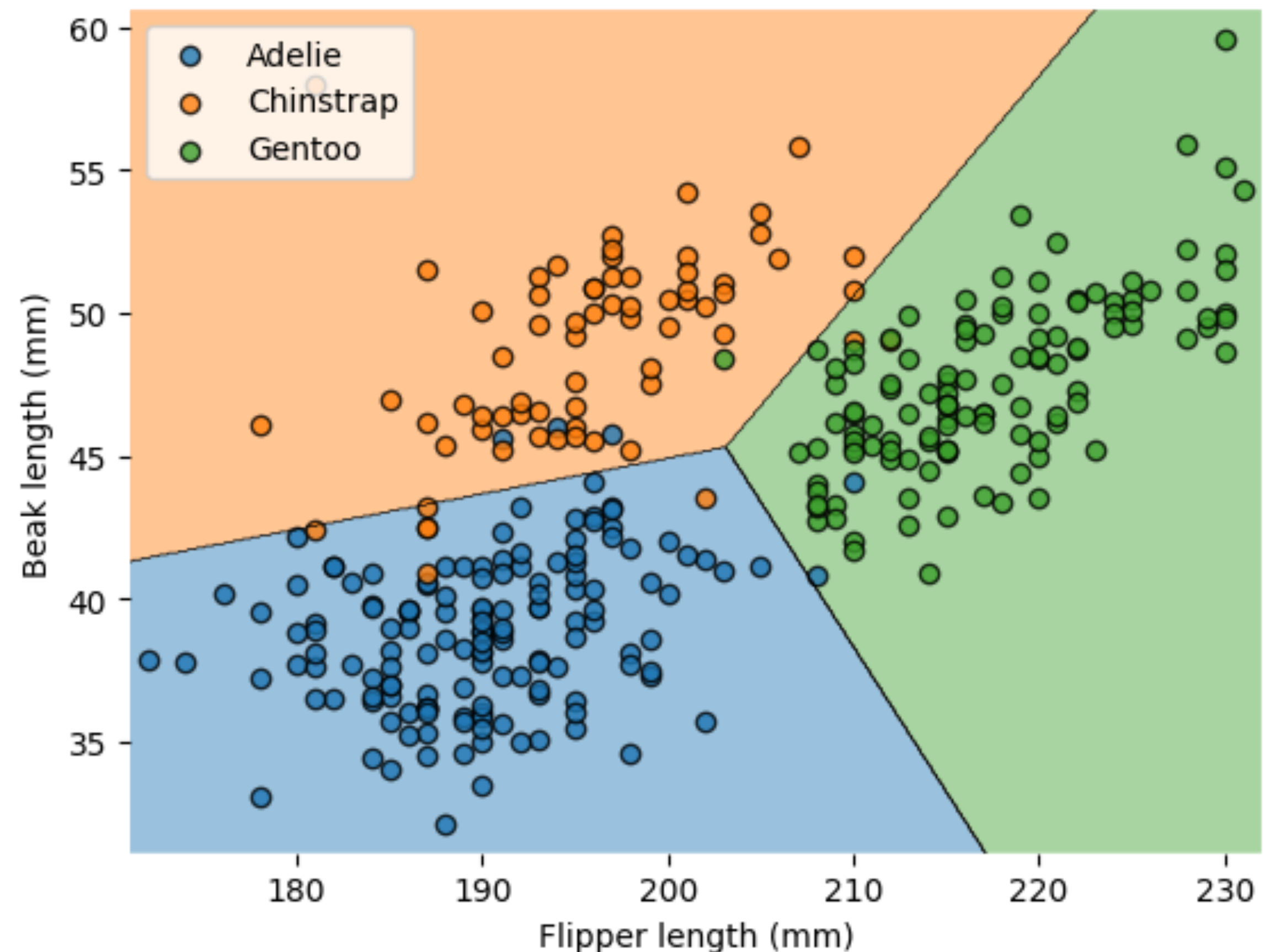
- Accuracy = # correct classifications / # total data points

96% accurate

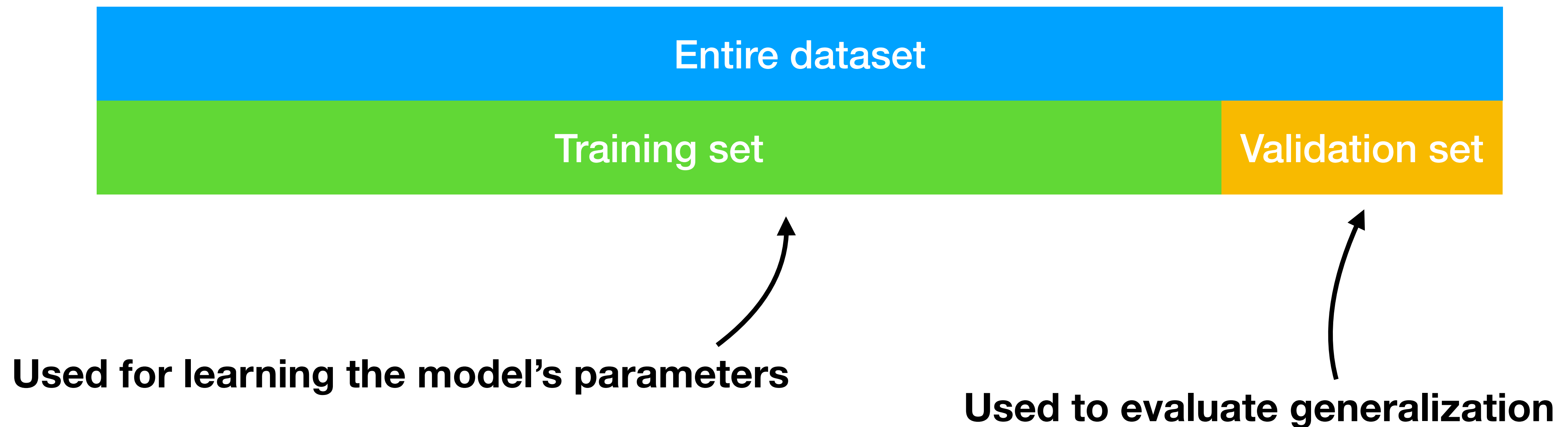


Evaluating model generalization

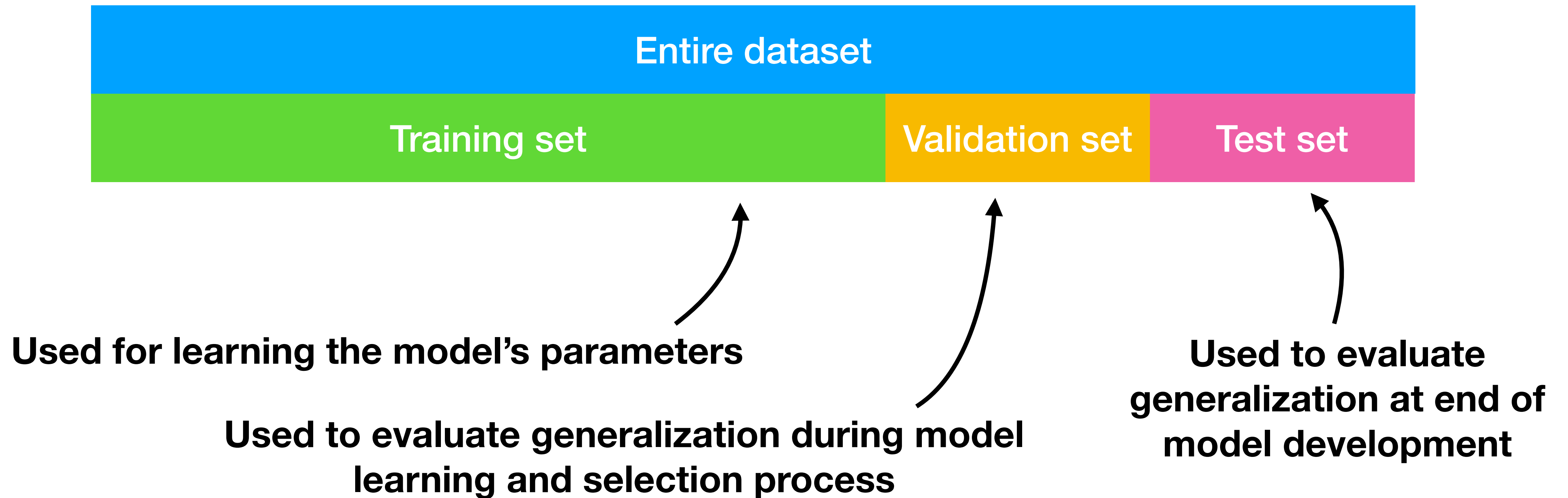
- So far we have fit and evaluated the model on the same data. Why might this not be ideal?
- Model generalization:
Performance of model on data **not** used for fitting / “training”



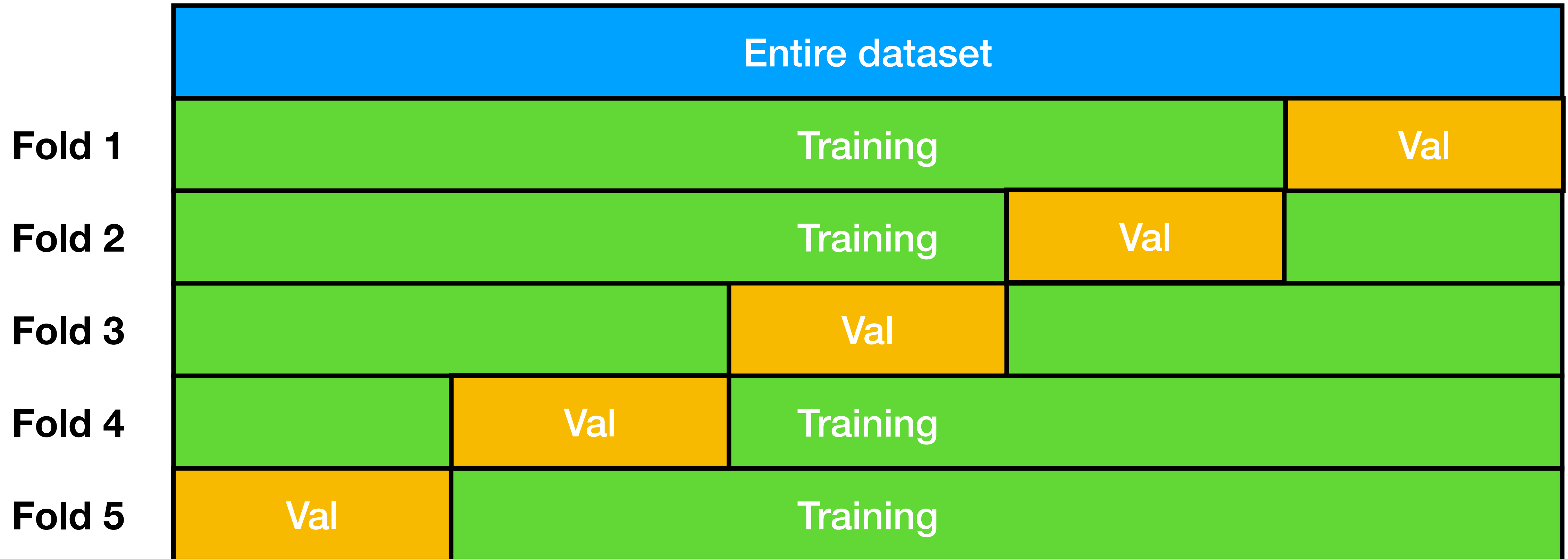
Data splits: training and validation



Test set



K-Fold Cross Validation



Compute average accuracy across all folds